

# SAMBP Unified Architecture — All Protection Functions

Layer 1: Physical Waveform Acquisition & CT Compensation

Layer 2: Forward Zone Model  $\theta \mapsto \hat{i}(t)$  [Reduced-parameter, bounded, analytical Jacobian]

Layer 3: Two-Pass Levenberg-Marquardt Inverse Estimator [ $\kappa_n$ , residual norm, confidence score]

Layer 4: Confidence Gate [conventional trip required;  $n_{confirm}$  hysteresis; blocking flags]

Layer 5: Protection Action [OC settings update / differential trip / mode-aware fallback]

**sambp\_xfmr**  
 $\theta = [I_{diff}, k_2, k_5, \varepsilon_{CT}]$   
87T Diff.

**sambp\_sync\_oc**  
 $\theta = [I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \varphi_a]$   
Overcurrent

adaptation / re-init

**sambp\_line**  
 $\theta = [I_{fund}, I_{DC}, \tau_{DC}]$   
87L Diff.